

Osteosarcoma Cell Nucleus Segmentation



CSE-4128: Image Processing and Computer Vision Laboratory

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1 Introduction

Microscopy images of stained tissue often suffer from low or uneven contrast, noise, and staining artifacts. Robust segmentation therefore benefits from (i) local contrast enhancement, (ii) denoising, and (iii) a simple, reproducible segmentation method like Otsu’s threshold optimized on the preprocessed image. We adopt CLAHE for local contrast control, Gaussian blur for gentle noise suppression, global Otsu thresholding, and standard morphology (opening/closing) to regularize masks before counting.

2 Methodology

Our pipeline segments osteosarcoma nuclei and computes region descriptors from brightfield images. It follows four major stages: preprocessing (contrast enhancement + denoising), global thresholding, morphological refinement, and component analysis with geometric descriptors [3].

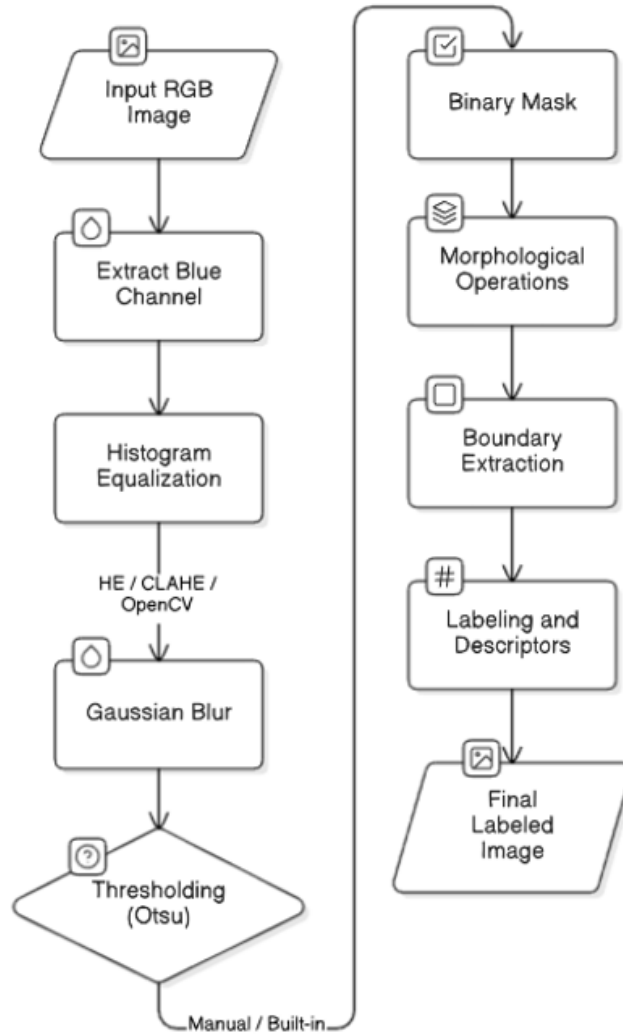


Figure 1: System Flow Diagram

Given an RGB input, we process:

1. **Blue channel extraction:** use the blue (hematoxylin-rich) channel as a grayscale proxy.
2. **Contrast enhancement:** apply CLAHE tile-wise with histogram clipping and redistribution.
3. **Smoothing:** convolve with a Gaussian kernel G_σ to suppress speckle while keeping edges.

4. **Thresholding:** global Otsu to obtain a binary mask.
5. **Morphology:** erosion/opening/dilation/closing to remove specks and fill small holes.
6. **Boundary Extraction:** Subtract eroded image from thresholded image to get the boundaries.
7. **Labeling:** 8-connected components to get per-nucleus labels.
8. **Measurement:** compute area, perimeter (from the border), and a max-diameter; then calculate compactness, form factor, and roundness for each nucleus.

2.1 Preprocessing

Channel selection. We first convert the RGB image to BGR (OpenCV convention) and extract the blue channel, which maximizes nucleus contrast against cytoplasm/background in DAPI-stained imagery.

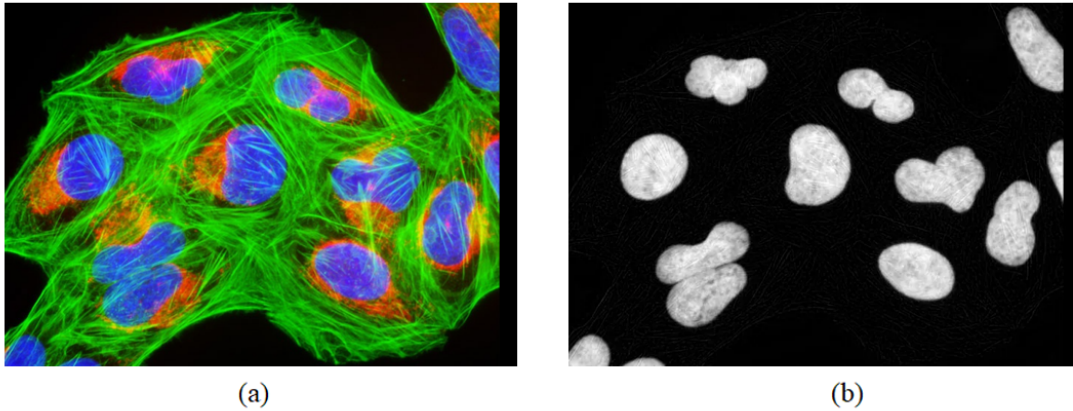


Figure 2: (a) Original Image (b) Blue channel extraction

Histogram Equalization Options. We expose three alternatives: (i) global histogram equalization (HE), (ii) manual CLAHE (our reference implementation), and (iii) OpenCV CLAHE.

2.1.1 Manual CLAHE

Adaptive histogram equalization (AHE) computes a local mapping per contextual region (tile); this improves local contrast but tends to amplify noise in homogeneous areas. **Clip-Limited adaptive Histogram Equalization (CLAHE)** locally gets the histogram and clips the tile histogram at a clip limit and redistributing the clipped mass uniformly across bins, thereby controlling noise amplification while retaining local contrast.

CLAHE divides the image into a grid of contextual tiles, equalizes each tile’s histogram, *clips* each histogram at a limit to prevent noise amplification, and *redistributes* the clipped mass uniformly before forming the cumulative distribution function (CDF) [1].

2.1.2 Algorithm

Let $L = 256$ gray levels. Given a tile T (size $t_h \times t_w$):

1. Compute histogram $h[0..L-1]$ and total count $|T|$.
2. Compute the per-bin clip threshold

$$h_{\max} = \text{clip_limit} \times \frac{|T|}{L}.$$

3. Clip: $\tilde{h}[i] = \min(h[i], h_{\max})$. Accumulate the excess

$$E = \sum_{i=0}^{L-1} (h[i] - h_{\max})_+,$$

where $(x)_+ = \max(x, 0)$.

4. Redistribute the excess uniformly:

$$\hat{h}[i] = \tilde{h}[i] + \frac{E}{L}.$$

5. Normalize to a PDF $p[i] = \hat{h}[i] / \sum_j \hat{h}[j]$ and compute CDF $c[i] = \sum_{k=0}^i p[k]$.

6. Build the lookup table $\text{LUT}[i] = \lfloor c[i](L-1) \rfloor$ and remap T by table lookup.

Notes on Clipping and “Excess” Clipping does *not* discard mass: the difference $(h[i] - h_{\max})_+$ is pooled as E and then evenly spread back, preserving the total histogram count and capping the local mapping slope. This is exactly what your implementation does with `excess = hist - max_clip`; `excess[excess<0]=0`; `hist = np.minimum(hist, max_clip)`; `hist += excess.sum()/L`. The redistribution maintains brightness on average while avoiding runaway amplification in peaked bins.

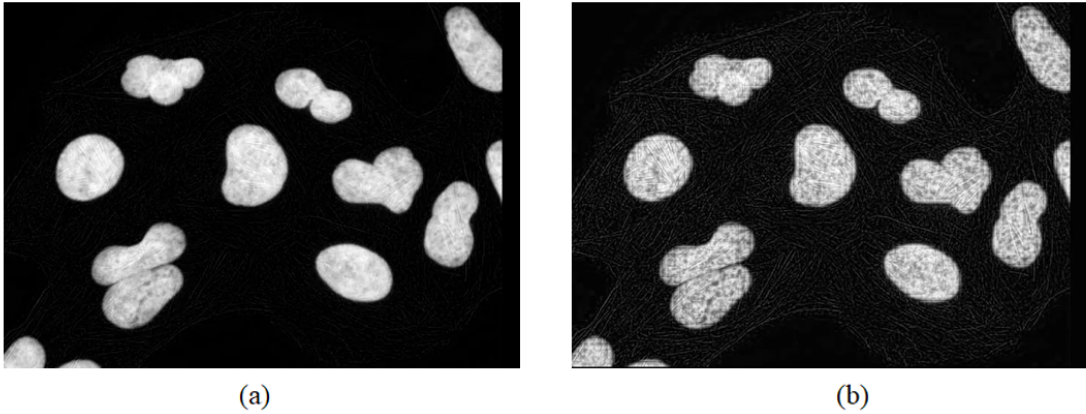


Figure 3: (a) Blue channel Extrated (b) Histogram Equalization (CLAHE)

2.1.3 Gaussian Denoising

Prior to thresholding, the image is smoothed using a **Gaussian filter** to attenuate high-frequency noise. The filtering operation is defined as a convolution with a Gaussian kernel

$$G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right),$$

where σ denotes the standard deviation controlling the degree of smoothing. Larger values of σ produce stronger blurring and greater noise suppression, while smaller values retain more fine-grained information. The parameter σ is user-tunable, allowing adaptive balance between noise reduction and edge preservation based on image quality.

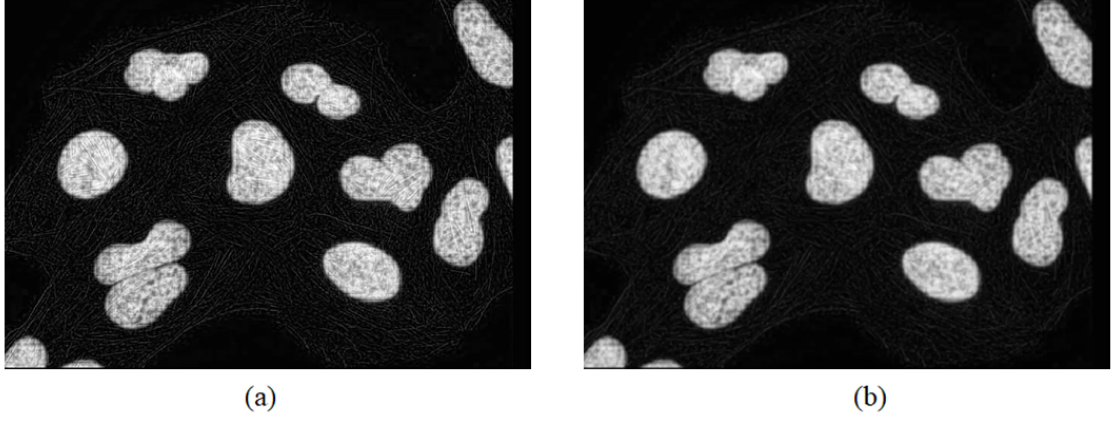


Figure 4: (a) Histogram Equalized (b) Gaussian Blurring

2.2 Segmentation via Thresholding

We adopt **Otsu's method** to find a single intensity threshold that maximizes class separability between foreground (nuclei) and background. Let L be the number of gray levels (256), p_i the normalized histogram, and $P_1(k) = \sum_{i=0}^k p_i$ the cumulative probability up to level k . Define the cumulative mean $m(k) = \sum_{i=0}^k i p_i$ and the global mean $m_G = \sum_{i=0}^{L-1} i p_i$. The between-class variance is

$$\sigma_B^2(k) = \frac{(m_G P_1(k) - m(k))^2}{P_1(k) (1 - P_1(k))}.$$

The Otsu threshold k^* is the gray level that maximizes $\sigma_B^2(k)$.

2.2.1 Algorithm

1. Compute normalized histogram p_i for $i=0, \dots, L-1$.
2. Compute cumulative sums $P_1(k)$ for $k=0, \dots, L-1$.
3. Compute cumulative means $m(k)$ for $k=0, \dots, L-1$.
4. Compute global mean m_G .
5. For each k , compute $\sigma_B^2(k)$; track the maximum value.
6. Set $k^* = k$ that has the maximum between-class variance.

Otsu is simple, fast, and effective when the histogram is near bimodal.

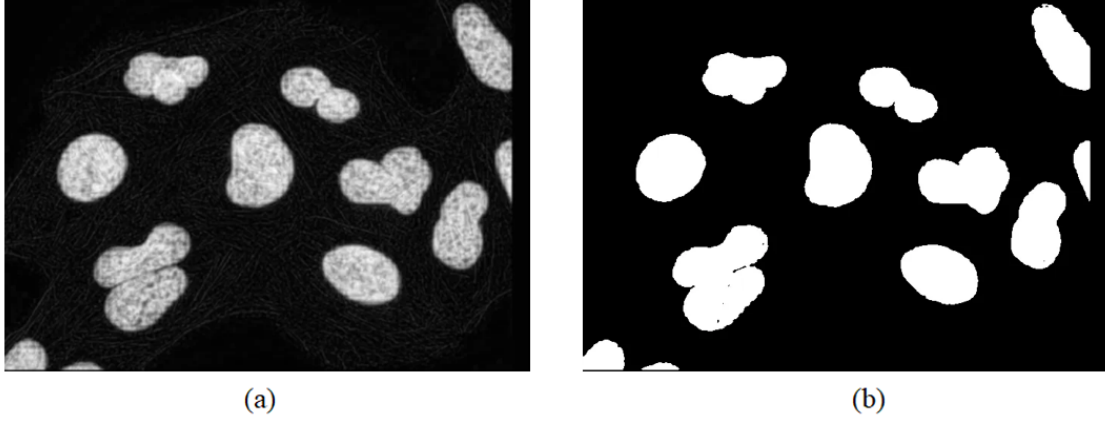


Figure 5: (a) Gaussian Blurring (b) Otsu's thresholding

2.3 Morphological Operation

Let B be a 3×3 structuring element of ones. We apply erosion ($\ominus$), dilation ($\oplus$), opening ($\circ$), and closing ($\bullet$) in a short sequence to remove salt noise, smooth ragged edges, and fill pinholes:

$$M_1 = I_b \circ B = (I_b \ominus B) \oplus B, \quad M_2 = M_1 \bullet B = (M_1 \oplus B) \ominus B.$$

The final binary mask M_2 is used for labeling.

2.4 Boundary Extraction

Let B be a 3×3 structuring element of ones. We apply erosion then subtract the eroded image from closed image to get the boundaries:

$$M_2 = M_2 - (M_2 \ominus B)$$

M_2 contains the boundaries for each nulkues

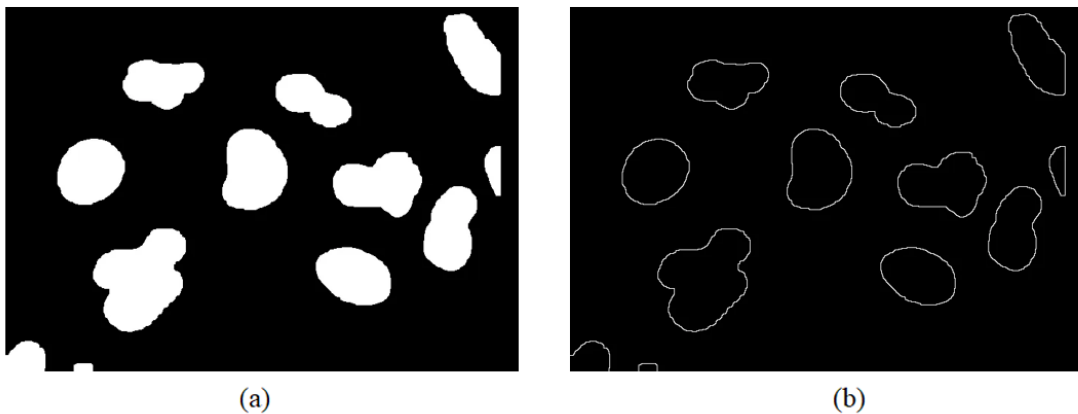


Figure 6: (a) Morphological operation (b) Boundary extraction

2.5 Connected-Component Labeling and Counting.

2.5.1 Algorithm:

1. Initialize label matrix $L(x,y)$ with zeros.

2. For each pixel (x, y) :
 - If $M_2(x, y) = 1$ (foreground) and $L(x, y) = 0$, assign a new label k .
 - Push (x, y) to a stack for region growing.
3. While the stack is not empty:
 - Pop a pixel (x', y') .
 - For each of its 8-connected neighbors (x_n, y_n) :
 - If $M_2(x_n, y_n) = 1$ and $L(x_n, y_n) = 0$, assign label k and push it to the stack.
4. Increment k and repeat until all pixels are visited.
5. The total nucleus count is $N = k - 1$.

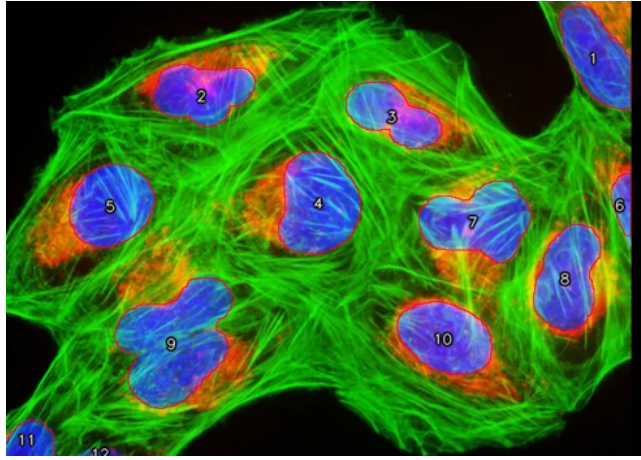


Figure 7: Nucleus Labeling and Counting

2.6 Region Descriptors.

1. Compute geometric properties:
 - **Area:** $A_k = \sum_{(x,y) \in \Omega_k} 1$.
 - **Perimeter:** $P_k = \sum_{(x,y) \in \partial\Omega_k} 1$.
 - **Maximum diameter:** $D_k = \max_{(x_1, y_1), (x_2, y_2) \in \Omega_k} ((x_1 - x_2), (y_1 - y_2))$.
2. Compute shape descriptors:

$$\begin{aligned}
 \text{Compactness: } C_k &= \frac{P_k^2}{A_k}, \\
 \text{Form Factor: } F_k &= \frac{4\pi A_k}{P_k^2}, \\
 \text{Roundness: } R_k &= \frac{4A_k}{\pi D_k^2}.
 \end{aligned}$$

3. Store all values in the descriptor set:

$$\mathcal{D} = \{(A_k, P_k, D_k, C_k, F_k, R_k)\}_{k=1}^N.$$

3 Software Components

3.1 Key Functions (Python/OpenCV/Numpy)

[leftmargin=2em]

- `manual_CLAHE`: tile histogram, clip to $h_{\max}$, redistribute E/L , LUT remap.
- `gaussian_kernel` & `apply_convolution`: explicit kernel build & reflective padding convolution.
- `manual_otsu_threshold`: histogram-based sweep to maximize between-class variance.
- `get_count_nuclei`: DFS/stack 8-connectivity to label components.
- `compute_region_descriptors`: for each label, compute A, P, D , then (F, R, C) .

3.2 UI and Outputs

The Gradio app provides:

1. Step-by-step visualization of the 10 stages (original $\rightarrow$ final).
2. Variant toggles for equalization (Manual HE, Manual CLAHE, OpenCV CLAHE) and thresholding (manual Otsu vs. OpenCV).
3. Final overlay with red boundaries and center labels plus a textual table of descriptors.

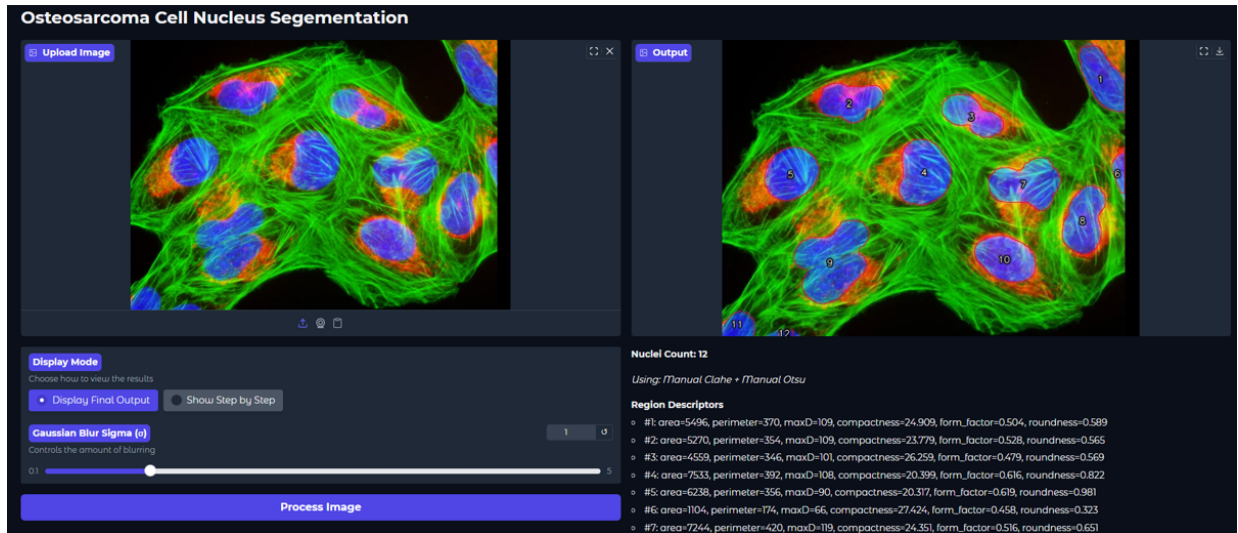


Figure 8: User Interface

4 Limitations

- **Otsu** assumes bimodality; very uneven illumination or non-bimodality reduces its efficiency [3].
- **CLAHE parameters** are image-dependent; excessive clipping can wash out contrast, too little reverts toward AHE and can amplify noise [1].

- Current morphological and threshold-based pipelines cannot fully separate overlapping or touching nuclei, which remain a major limitation.

Future work should integrate watershed, contour evolution, or graph-based splitting methods to isolate these overlaps [3].

5 Discussion

The project demonstrates that traditional, interpretable image processing techniques remain effective for nuclei segmentation in osteosarcoma microscopy images. By combining manual Contrast-Limited Adaptive Histogram Equalization (CLAHE), Gaussian smoothing, and Otsu thresholding, the approach achieves robust separation of nuclei from the cytoplasmic background under varying illumination conditions.

The manual CLAHE implementation plays a central role in mitigating uneven lighting and low contrast typical of histopathological images. Its tile-based adaptive enhancement ensures local brightness normalization while constraining over-amplification of noise through clipping. This yields better contrast enhancement compared to global histogram equalization, consistent with prior findings in medical imaging enhancement studies [1].

Gaussian denoising prior to thresholding significantly reduces high-frequency noise, stabilizing Otsu’s algorithm. However, the method’s assumption of bimodality still poses challenges when nuclei produce multiple peaks in the histogram. Morphological operations such as erosion, dilation, opening, and closing effectively refine the binary masks—removing small noise and filling holes—but are not sufficient for separating touching nuclei.

Connected-component labeling offers an efficient counting mechanism for distinct nuclei regions. However, the system does not yet handle overlapping cells limiting its scope.

6 Conclusion

This work presents an end-to-end, interpretable, and modular pipeline for osteosarcoma nuclei segmentation using purely classical image processing methods. The integration of manual CLAHE for adaptive contrast enhancement, Gaussian denoising, and Otsu thresholding provides a balance between computational efficiency and visual quality. The method achieves reliable nuclei counting and region-level descriptor extraction.

Despite its simplicity, the system exhibits limitations due to global thresholding and morphological segmentation — including inability to resolve overlapping nuclei.

Future work may involve extending the framework with adaptive watershed splitting, gradient-based boundary refinement, and texture or deep feature descriptors for improved robustness and discriminative analysis. .

References

- [1] K. Zuiderveld. (1994). *Contrast Limited Adaptive Histogram Equalization (CLAHE)*.
- [2] J. Lee et al. (2015). *An Adaptive Histogram Equalization Based Local Technique for Contrast Preserving Image Enhancement*.
- [3] K. Y. Win, S. Choomchuay, K. Hamamoto, and M. Raveesunthornkiat, “Comparative study on automated cell nuclei segmentation methods for cytology pleural effusion images,” *Journal of Healthcare Engineering*, vol. 2018